

Bootstrap — Topics 19-22: Cards, Navbar, Dropdowns & List Group

Study paper format: notes (Roman Urdu + English technical terms) + active recall Q&A at the end. Covers: Cards (★ ★ ★), Navbar (★ ★ ★), Dropdowns, List Group.

19. Cards ★ ★ ★

Card ek **flexible content container** hai — border, padding, aur options ke sath. Almost har modern UI (product listings, profiles, blog previews) mein cards use hoti hain.

Basic Card

```
html

<div class="card" style="width: 18rem;">
  <div class="card-body">
    <h5 class="card-title">Card Title</h5>
    <p class="card-text">Some quick example text for the card.</p>
    <a href="#" class="btn btn-primary">Go somewhere</a>
  </div>
</div>
```

- `.card` → outer wrapper (border + rounded corners)
- `.card-body` → padding wala content area
- `.card-title` → heading style
- `.card-text` → paragraph style

Card Image

```
html
```

```
<div class="card" style="width: 18rem;">
  
  <div class="card-body">
    <h5 class="card-title">Card with Image</h5>
    <p class="card-text">Image ke neeche content.</p>
  </div>
</div>
```

`.card-img-top` image ko card ke top par fit karti hai (rounded corners top par match karti hai).

Variants: `.card-img-bottom` (image neeche), ya `.card-img` (poore card ka background bana deti hai).

Card Body

`.card-body` andar aata hai jab bhi actual padded content chahiye ho — title, text, links, buttons sab isi ke andar hote hain.

```
html

<div class="card-body">
  <h5 class="card-title">Title</h5>
  <h6 class="card-subtitle mb-2 text-muted">Subtitle</h6>
  <p class="card-text">Text content.</p>
  <a href="#" class="card-link">Card Link</a>
</div>
```

`.card-subtitle` aur `.card-link` bhi available hain extra styling ke liye.

Card Footer (aur Header)

```
html
```

```
<div class="card">
  <div class="card-header">
    Featured
  </div>
  <div class="card-body">
    <h5 class="card-title">Special Title</h5>
    <p class="card-text">Content with a header and footer.</p>
  </div>
  <div class="card-footer text-muted">
    2 days ago
  </div>
</div>
```

`.card-header` top par ek separated section deta hai, `.card-footer` bottom par — dono `.card-body` se background-wise separate dikhte hain (light gray divider).

Card Groups

Multiple cards ko **equal height aur connected** border ke sath side-by-side render karta hai.

```
html

<div class="card-group">
  <div class="card">...</div>
  <div class="card">...</div>
  <div class="card">...</div>
</div>
```

`.card-group` cards ke beech koi gap nahi rakhta aur unki height automatically equal ho jati hai.

Compare: Agar gap wale equal-width cards chahiye (grid jaisa), to `.card-group` ki bajaye Grid System (`row` + `col`) use karna zyada common aur flexible hota hai:

html

```
<div class="row row-cols-1 row-cols-md-3 g-4">
  <div class="col"><div class="card">...</div></div>
  <div class="col"><div class="card">...</div></div>
  <div class="col"><div class="card">...</div></div>
</div>
```

20. Navbar ★ ★ ★

Navbar website ki navigation bar hai — responsive, collapsible, aur bohat customizable.

Brand

html

```
<a class="navbar-brand" href="#">MyBrand</a>
```

`.navbar-brand` logo/site-name ke liye — bold aur bari text milti hai.

Links

html

```
<ul class="navbar-nav">
  <li class="nav-item">
    <a class="nav-link active" href="#">Home</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">About</a>
  </li>
</ul>
```

`.navbar-nav` links ki list hai, har link `.nav-item` wrapper mein `.nav-link` hoti hai.
`.active` class current page ko highlight karti hai.

Collapse

```
html

<div class="collapse navbar-collapse" id="navbarNav">
  <ul class="navbar-nav">
    ...
  </ul>
</div>
```

`.navbar-collapse` navbar content ko wrap karta hai jo chhoti screens par **hide/collapse** ho jata hai (hamburger menu ke peeche).

Toggler

```
html

<button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-b
  <span class="navbar-toggler-icon"></span>
</button>
```

`.navbar-toggler` wo hamburger icon button hai jo mobile par dikhta hai — click karne par `.navbar-collapse` wale content ko show/hide karta hai (`data-bs-toggle` aur `data-bs-target` isi ko control karte hain, koi custom JS ki zaroorat nahi).

Responsive Navbar (Complete Example)

```
html
```

```

<nav class="navbar navbar-expand-lg navbar-light bg-light">
  <div class="container-fluid">
    <a class="navbar-brand" href="#">MyBrand</a>

    <button class="navbar-toggler" type="button" data-bs-toggle="collapse"
      <span class="navbar-toggler-icon"></span>
    </button>

    <div class="collapse navbar-collapse" id="navbarNav">
      <ul class="navbar-nav ms-auto">
        <li class="nav-item"><a class="nav-link active" href="#">Home</a>
        <li class="nav-item"><a class="nav-link" href="#">About</a></li>
        <li class="nav-item"><a class="nav-link" href="#">Contact</a>
      </ul>
    </div>
  </div>
</nav>

```

- `navbar-expand-lg` → ≥ 992 px se navbar horizontally expand ho jata hai (links show), usse choti screens par collapsed/hamburger mode mein rehta hai.
- `navbar-light` / `navbar-dark` → text color light/dark background ke hisab se set karti hai.
- `ms-auto` (margin-start auto) → links ko right side push kar deta hai.

Breakpoint variants: `navbar-expand-sm`, `navbar-expand-md`, `navbar-expand-lg`, `navbar-expand-xl`, `navbar-expand-xxl` — jitni choti value utni jaldi navbar expand (links show) ho jata hai.

Dropdown Inside Navbar

html

```

<li class="nav-item dropdown">
  <a class="nav-link dropdown-toggle" href="#" role="button" data-bs-toggle="dropdown">
    Services
  </a>
  <ul class="dropdown-menu">
    <li><a class="dropdown-item" href="#">Web Design</a></li>
    <li><a class="dropdown-item" href="#">SEO</a></li>
  </ul>
</li>

```

Navbar ke andar `.dropdown` wrapper, `.dropdown-toggle` link, aur `.dropdown-menu` list se ek dropdown navbar item ban jata hai.

21. Dropdowns

Dropdown ek clickable trigger hota hai jo ek hidden menu show karta hai.

Basic Dropdown

```

html

<div class="dropdown">
  <button class="btn btn-primary dropdown-toggle" type="button" data-bs-toggle="dropdown">
    Dropdown Button
  </button>
  <ul class="dropdown-menu">
    <li><a class="dropdown-item" href="#">Action</a></li>
    <li><a class="dropdown-item" href="#">Another action</a></li>
    <li><hr class="dropdown-divider"></li>
    <li><a class="dropdown-item" href="#">Something else</a></li>
  </ul>
</div>

```

`.dropdown-toggle` trigger button/link hai, `.dropdown-menu` menu container, `.dropdown-item` har option, aur `.dropdown-divider` ek horizontal separator line.

Split Dropdown

Do separate buttons — ek normal action ke liye, doosra sirf dropdown open karne ke liye.

```
html

<div class="btn-group">
  <button type="button" class="btn btn-primary">Save</button>
  <button type="button" class="btn btn-primary dropdown-toggle dropdown-tog
  </button>
  <ul class="dropdown-menu">
    <li><a class="dropdown-item" href="#">Save as draft</a></li>
    <li><a class="dropdown-item" href="#">Save and publish</a></li>
  </ul>
</div>
```

`dropdown-toggle-split` chota arrow-only button banata hai jo main action button se attached hota hai.

Dropup, Dropstart, Dropend

Menu ki direction change karne ke liye wrapper class change karo:

```
html

<div class="dropup">...</div>      → menu upar khulta hai
<div class="dropstart">...</div>   → menu left side par khulta hai
<div class="dropend">...</div>     → menu right side par khulta hai
```

(Default `.dropdown` se menu neech khulta hai.)

22. List Group

List Group flexible list of items dikhane ke liye hota hai — simple lists se le kar complex interactive lists tak.

Basic List Group

```
html

<ul class="list-group">
  <li class="list-group-item">Item 1</li>
  <li class="list-group-item">Item 2</li>
  <li class="list-group-item">Item 3</li>
</ul>
```

Active

```
html

<li class="list-group-item active">Currently selected item</li>
```

Item ko highlighted/selected style deta hai (blue background — theme ke hisab se).

Disabled

```
html

<li class="list-group-item disabled">Unavailable item</li>
```

Item ko grayed-out aur non-interactive dikhata hai.

Flush

```
html
```

```
<ul class="list-group list-group-flush">
  <li class="list-group-item">Item 1</li>
  <li class="list-group-item">Item 2</li>
</ul>
```

`.list-group-flush` list ke rounded corners aur outer border remove kar deta hai — cards ke andar embed karne ke liye best (seamless look).

Numbered

```
html

<ol class="list-group list-group-numbered">
  <li class="list-group-item">First item</li>
  <li class="list-group-item">Second item</li>
</ol>
```

`.list-group-numbered` automatically har item ke sath number prefix add kar deta hai (`<ol>` tag ke sath).

Horizontal

```
html

<ul class="list-group list-group-horizontal">
  <li class="list-group-item">Item 1</li>
  <li class="list-group-item">Item 2</li>
  <li class="list-group-item">Item 3</li>
</ul>
```

Items vertically stack hone ki bajaye horizontally side-by-side arrange hote hain.

Responsive variant: `list-group-horizontal-md` (sirf $\geq 768\text{px}$ se horizontal, usse choti screens par vertical).

Cheat Sheet

Section	Key Classes
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Card	<code>card</code> , <code>card-body</code> , <code>card-img-top</code> , <code>card-title</code> , <code>card-text</code> , <code>card-header</code> , <code>card-footer</code> , <code>card-group</code>
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Navbar	<code>navbar</code> , <code>navbar-brand</code> , <code>navbar-nav</code> , <code>nav-item</code> , <code>nav-link</code> , <code>navbar-collapse</code> , <code>navbar-toggler</code> , <code>navbar-expand-*</code>
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Dropdown	<code>dropdown</code> , <code>dropdown-toggle</code> , <code>dropdown-menu</code> , <code>dropdown-item</code> , <code>dropdown-toggle-split</code> , <code>dropup</code> , <code>dropstart</code> , <code>dropend</code>
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List Group	<code>list-group</code> , <code>list-group-item</code> , <code>active</code> , <code>disabled</code> , <code>list-group-flush</code> , <code>list-group-numbered</code> , <code>list-group-horizontal</code>
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Practice Exercises

1. Ek product card banao jisme `card-img-top`, `card-body` (title + text + price), aur `card-footer` ("Add to Cart" button) ho.
2. Ek complete responsive navbar banao — brand, 3 links, aur ek dropdown ("Services") jisme 2 options hon. `navbar-expand-lg` use karo.
3. Ek split dropdown button banao jisme main action "Download" ho aur dropdown mein "Download as PDF" / "Download as CSV" options hon.
4. Ek `list-group-numbered` banao jisme 5 steps ho, aur ek item ko `active` mark karo.
5. Card ke andar `list-group-flush` embed karo (jaise pricing features list ek card ke andar).

Active Recall — Cards, Navbar, Dropdowns & List Group

┆ Khud se answer karne ki koshish karo, phir neeche scroll karke check karo.

- Q1. Card ke basic structure mein `.card-body` ka kya role hai?
- Q2. `.card-img-top` aur `.card-img-bottom` mein kya farq hai?
- Q3. `.card-group` kya karta hai, aur ye grid-based card layout (`row` + `col`) se kaise different hai?
- Q4. Navbar mein `.navbar-toggler` aur `.navbar-collapse` ka combined role kya hai?
- Q5. `navbar-expand-lg` ka matlab kya hai — kis breakpoint se navbar expand hoga?
- Q6. `ms-auto` navbar links ke context mein kya karta hai?
- Q7. Split dropdown normal dropdown se kaise different hai?
- Q8. `dropdown`, `dropstart`, aur `dropend` classes kya control karti hain?
- Q9. `list-group-flush` kab use karte hain, aur ye kya remove karta hai?
- Q10. `list-group-numbered` ke liye konsa HTML tag use karna zaroori hai (`<ul>` ya `<ol>`)?
- Q11. `list-group-horizontal` default list group se kaise different hai?
-

Answers

<details> <summary>Answers dekhne ke liye click karein</summary>

- A1. `.card-body` card ke andar ka padded content area hai — title, text, links, buttons sab isi ke andar aate hain, taake unhe proper spacing mile.
- A2. `.card-img-top` image ko card ke sabse upar (title/text se pehle) fit karti hai, jabke `.card-img-bottom` image ko card ke sabse neeche (content ke baad) fit karti hai.
- A3. `.card-group` multiple cards ko bina gap ke connected border ke sath equal height mein render karta hai. Grid-based approach (`row` + `col`) cards ke beech gap allow karta hai aur zyada flexible/responsive control deta hai (kitne cards per row har breakpoint par) — isliye grid approach zyada commonly use hoti hai.

A4. `.navbar-toggler` wo hamburger button hai jo click hone par `.navbar-collapse` wale content (links, dropdowns) ko show ya hide karta hai — chhoti screens par navbar ko collapsible banane ke liye ye dono sath kaam karte hain.

A5. `navbar-expand-lg` ka matlab hai ke $\geq 992\text{px}$ (large breakpoint) se navbar horizontally expand ho jayega (links directly visible honge); usse choti screens par navbar collapsed/hamburger mode mein rahega.

A6. `ms-auto` (margin-start: auto) navbar links wali `<ul>` ko available space mein **right side push** kar deta hai — isse brand left par aur links right par align hote hain.

A7. Split dropdown mein do separate clickable buttons hote hain — ek main action perform karta hai (jaise "Save"), aur doosra chota arrow-only button sirf dropdown menu open karta hai. Normal dropdown mein pura button/link hi trigger hota hai dropdown open karne ke liye.

A8. Ye classes dropdown menu ki **opening direction** control karti hain — `dropup` menu ko upar khulwati hai, `dropstart` left side, aur `dropend` right side (default `.dropdown` neeche khulta hai).

A9. `list-group-flush` tab use karte hain jab list ko kisi card ya container ke andar seamlessly embed karna ho. Ye list ke rounded corners aur outer border ko remove kar deta hai.

A10. `list-group-numbered` ke sath `<ol>` (ordered list) tag use karna zaroori hai, taake automatic numbering theek se kaam kare.

A11. `list-group-horizontal` mein items vertically stack (upar-neeche) hone ki bajaye horizontally side-by-side arrange hote hain.

`</details>`